

Subject: Application for Working Student Position in Robotics

Dear Hiring Team,

I am currently pursuing my Master's degree in Electrical and Information Technology at Hochschule Darmstadt, specializing in Automation and Robotics. I am completing a mandatory internship at Bär Automation GmbH in Stuttgart and will have completed 80 ECTS credits by May 2026, after which I am available to start a working student position at BEC Robotics. Your focus on programming industrial robots, integrating sensors, and developing AI-driven automation solutions aligns perfectly with my academic background and professional interests.

During my internship as a Mobile Robots Development Intern at Bär Automation GmbH within the ARENA2036 research environment in Stuttgart, I gained practical experience working with autonomous mobile robots and their navigation systems. Using ROS 2, RVIZ, Gazebo and the Nav2 stack, I explored and compared SLAM approaches such as SLAM Toolbox and Google Cartographer by generating and evaluating maps under different configurations. I also worked on improving robot localization by tuning LiDAR scan matching and odometry parameters to reduce positional drift and improve navigation accuracy. In addition, I became familiar with the VDA 5050 standard used for communication between AGVs and fleet management systems. As part of the same environment, I also contributed to the development of digital twin tools by helping build a Python and React-based tool that automatically generates Asset Administration Shells (AAS v3.1) for industrial assets. This work gave me exposure to structured data models and the integration of physical systems with their digital representations.

In addition to my internship, I gained experience with robotic perception systems through a university project where my team implemented a mapping and perception pipeline in Python and ROS for a UniTree quadruped robot equipped with a Hesai XT16 LiDAR sensor. This project involved processing point cloud data and generating indoor maps using filtering and segmentation techniques. Through this work, I developed a strong practical understanding of how robots interpret spatial data and convert it into representations that support environment understanding and scene interpretation. This background relates closely to the challenges described in your thesis position. Working directly with LiDAR data, mapping pipelines, and ROS 2 has equipped me to tackle complex sensor data, evaluate unexpected behavioral patterns, and navigate simulated environments. This involved processing point clouds, segmenting indoor spaces, and generating accurate maps directly relevant to developing intelligent robot perception and motion planning systems at BEC Robotics.

My previous experience as a Senior System Engineer at Infosys provided me with a strong foundation in programming, Agile development processes, and cross-functional collaboration. Combined with my robotics experience, I am confident in my ability to contribute effectively to your projects, whether programming KUKA robots, teaching precise motion sequences, integrating sensors, or developing modern software interfaces for industrial automation.

I am excited about the opportunity to join BEC Robotics, contribute to real-world robotics applications, and expand my expertise in automation technologies. My availability, flexible working style, and enthusiasm for robotics make me a motivated candidate to support your team's innovative projects.

Thank you for considering my application. I look forward to the possibility of discussing how I can contribute to your team.

Best regards,
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